

Valid covariance functions on circulant lattices, and what the FFT algebra actually requires

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1 Setting and the reviewers' point

The manuscript builds stationary correlation matrices on a cyclic lattice by composing a Euclidean correlation function with the great-circle (wrap-around) distance: with $h_{ik} = \min(|i - k|, n - |i - k|)$,

$$R_{ik} = \rho(h_{ik}), \quad \rho(h) = \exp\{-(h/\phi)^p\}, \quad p \in (0, 2], \quad (1)$$

(eq. (correlation) in the paper, with “ $p = 2$ induces smooth realizations”). The reviewers correctly note that (1) is not a valid construction in general: a function that is positive definite (PD) on $\mathbb{R}$ does not remain PD after composition with the geodesic distance on a circle. In the experiments this was handled by keeping ϕ small enough that the matrix was numerically SPD. This note summarizes the relevant theory, checks which published experiments were affected (none, it turns out), and proposes concrete modifications.

2 When is a circulant correlation matrix valid?

On the cyclic group $\mathbb{Z}_n$ the answer is exact and cheap to check. A stationary correlation on $\mathbb{Z}_n$ is a circulant matrix R with base $r = (r_0, \dots, r_{n-1})$, $r_h = \rho(h)$, and its eigenvalues are the DFT of the base. The discrete analogue of Bochner's theorem (the Herglotz theorem; see, e.g., Brockwell and Davis, 1991, Ch. 4) states:

R is a valid (positive semidefinite) covariance on $\mathbb{Z}_n$ *iff* $\text{DFT}(r)_k \geq 0$ for all $k = 0, \dots, n-1$.

Since the code already computes $\text{DFT}(r)$ to do FFT algebra, validity can and should be asserted at construction time for free.

3 Four provably valid constructions

(a) Restrict the smoothness exponent: $p \in (0, 1]$. Gneiting (2013) (building on Schoenberg, 1942) characterizes the isotropic PD functions on spheres S^d ; the circle S^1 contains every $\mathbb{Z}_n$ as a subset, so PD on S^1 implies PD on $\mathbb{Z}_n$ with great-circle distance. The powered exponential $\exp\{-(h/\phi)^p\}$ is PD on all spheres *iff* $p \in (0, 1]$, and the geodesic Matérn is PD *iff* $\nu \leq 1/2$. This explains the numerical table below: the paper's $p = 1$ choices (image and noise correlations) are valid *at every range* ϕ , whereas $p = 1.98$ (blur prior) is not a valid model on the circle and fails numerically once ϕ is large relative to n .

(b) Periodize (wrap) the Euclidean covariance. For any PD, integrable C on $\mathbb{R}$, the periodization

$$C_{\text{per}}(h) = \sum_{j \in \mathbb{Z}} C(h + jn) \quad (2)$$

is PD on $\mathbb{Z}_n$: by Poisson summation, DFT of the wrapped base samples the (nonnegative) spectral density of C . This is the classical device behind circulant embedding (Wood and Chan, 1994; Dietrich and Newsam, 1997; Stein, 1999). It preserves the Euclidean family’s smoothness, so $p \rightarrow 2$ stays available. In practice two or three terms of (2) suffice (C decays fast), and the change to `exp_cor` is one line: $r_h = \sum_{j=-J}^J \exp\{-(|h + jn|/\phi)^p\}$.

(c) Chordal distance. Composing any PD radial function on $\mathbb{R}^2$ with the *chordal* distance $c(h) = \frac{n}{\pi} \sin(\pi h/n)$ (the straight-line distance between lattice points placed on a circle of circumference n) is PD (Yadrenko, 1983; Gneiting, 2013). This keeps the closed form $\exp\{-(c(h)/\phi)^p\}$ valid for all $p \in (0, 2]$; chordal and great-circle distances agree to first order for $h \ll n$, so fitted ranges barely change.

(d) Compact support within the half-period. If C is PD on $\mathbb{R}$ and supported on $[0, n/2]$ (e.g. Wendland or other compactly supported families; Wendland, 1995; Gneiting, 2002), then (2) has a single term and the naive construction (1) is *exactly* valid. This is the rigorous version of the “keep ϕ small” workaround: for the powered exponential the support is infinite, so a small ϕ only makes the negative part of the spectrum exponentially small, never zero (for $p \approx 2$ the worst eigenvalue behaves like $-e^{-(n/2\phi)^2}$ -ish; see the table). An alternative in the same spirit is the Gaussian Markov random field route: specify the *precision* directly on the torus (Rue and Held, 2005), which is valid by construction and keeps the circulant structure.

4 Are the published experiments affected?

Minimum circulant eigenvalue of the naive base (1) and of the wrapped base (2):

n	p	ϕ	min eig, naive	min eig, wrapped
24	1.98	1.5	$2.7 \cdot 10^{-2}$	$2.7 \cdot 10^{-2}$
24	1.98	2.0	$3.6 \cdot 10^{-3}$	$3.6 \cdot 10^{-3}$
24	1.98	6.0	$-1.5 \cdot 10^{-2}$	$2.6 \cdot 10^{-4}$
24	1.98	10.0	$-3.7 \cdot 10^{-1}$	$9.1 \cdot 10^{-5}$
24	1.00	10.0	$3.5 \cdot 10^{-2}$	$5.0 \cdot 10^{-2}$
96	1.98	10.0	$9.1 \cdot 10^{-5}$	$9.1 \cdot 10^{-5}$

The experiments use $p = 1$ with $\phi = 1.5$ for image and noise (valid by (a) at any range) and $p = 1.98$ with $\phi \in \{1.5, 2\}$ for the blur prior on $n_v \in \{24, \dots\}$, where the naive minimum eigenvalue is $\geq 3.6 \cdot 10^{-3} > 0$. **All published experiments therefore used genuinely SPD matrices**; the issue is that the *recipe* stated in the paper is not valid in general, and a referee can break it with $p = 2$ and a larger ϕ .

5 Suggested modifications to the paper and code

1. **Eq. (correlation) (Sec. “Blur prior distribution”)**. State the validity condition once: “a stationary correlation on the cyclic lattice is valid iff the DFT of its base is nonnegative (Herglotz)”. Then present the construction as either (a) $p \in (0, 1]$ with great-circle distance (Gneiting, 2013), or (b) the periodization (2) of the Euclidean powered exponential for $p \in (0, 2]$, which we recommend since the blur prior wants $p \approx 2$ smoothness. Drop “choosing $p = 2$ induces smooth realizations” in favour of the wrapped version.
2. **SM simulation details**. Note that the values used ($p = 1.98, \phi \leq 2$; $p = 1, \phi = 1.5$) give SPD matrices, so no published result changes; for general use the wrapped form should be the default.
3. **Code**. In `src/utils/model_utils.py::exp_cor` (circulant branch): replace the base by the J -term wrapped sum (b), and assert `np.fft.fft(r).real.min() >= -tol` in `Covariance.create_exp_cor_matrix` so an invalid choice fails loudly instead of corrupting a chain.

6 Separability and the Kronecker product (Reviewer 3, comment 5)

The reviewer asks why the image covariance is assumed separable ($\mathbf{R}_c = \mathbf{R}_h \otimes \mathbf{R}_v$), suggesting the assumption is restrictive and perhaps unnecessary. The precise situation, verified against the implementation:

What the FFT speedups actually need: stationarity, not separability. Every matrix in the collapsed likelihood ($A = W\Sigma_c W^\top + \Sigma_d$, Z , B , their inverses, determinants and mat-vecs) is manipulated only through the 2D DFT of its base. Any *stationary* covariance on the torus yields a BCCB matrix, which the DFT diagonalizes whether or not it factorizes as a Kronecker product. The same holds for the Gibbs full conditionals and for the exact-observation correction (the gather $A_c C A_c^\top$ of a base column works for any BCCB). So the reviewer is right that the headline $O(n \log n)$ complexity does not require separability: a nonseparable wrapped 2D Matérn would drop straight into most of the algebra.

Where separability *is* used. Three places, all verified in the code: (i) the model is *specified* through one-dimensional ranges (ρ_v, ρ_h) , i.e. separability is a modeling convenience for building $\mathbf{R}_c$ from 1D pieces; (ii) the constrained-image factorizations: after conditioning on a full well column, the conditional covariance of a separable field factorizes, and the stored $\mathbf{R}_v^*, \mathbf{R}_h^*$ (and the Cholesky factor L) exploit this; (iii) the sum-of-squares terms of the potential gradient use the factorization explicitly ($K = P\mathbf{R}_h P^\top$, $X = \mathbf{R}_v^* W_0^\top$, and the $\mathbf{R}_{cv}, \mathbf{R}_{ch}$ eigenvalue products in the legacy gradient).

Suggested response / rewrite. Introduce stationarity on the extended cyclic lattice as the essential assumption (with the validity condition of Sec. 2) when the model is first presented, and present separability as an explicit, secondary assumption adopted (i) to elicit the prior through directional ranges, and (ii) to obtain closed-form factorizations in the exact-observation algebra

and the HMC gradient. It is fair to concede that the collapsed likelihood itself, and hence Gibbs and the HMC potential, extend verbatim to nonseparable stationary covariances; the gradient's sum-of-squares terms would need a (derivable) generalization. In short: the reviewer's instinct is correct for the core algebra and the paper overstated the role of the Kronecker product there, but separability is not vacuous either — it is what makes the well-constraint and gradient terms cheap in the current implementation.

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